Stochastic Analysis

LectureNotes

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Chapter 1

Intro

Riemann-Stieltjes Integral: If f is continuous and g is of finite variation then $\int_a^b f(x)\ \mathrm{d}g(x)$ is well-defined and is defined to be

$$\int_a^b f(x)\ \mathrm{d}g(x) := \lim_{n\to\infty} \sum_{i=0}^{n-1} f(\xi_i)\left(g(x_{i+1}) - g(x_i)\right)$$

with $a \le x_0 < x_1 < \cdots < x_n \le b$ and $\xi_i \in [x_i, x_{i+1}]$.

Main Task: Itô integral $(H \cdot X)_t = \int_0^t H_s\ \mathrm{d}X_s$.

E.g X is Brownian motion.

Theorem 1.0.1. $\displaystyle\sum_{k=1}^{2^n} \left(B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}}\right)^2 \xrightarrow{a.s.} 1.$

Finite Variation of a function $g : [a, b] \to \mathbb{R}$ is

$$V_g([a, b]) = \int_a^b |\mathrm{d}g(x)| = \sup_n \sum_{i=1}^{n} |g(x_{i+1}) - g(x_i)|$$

Corollary 1.0.1. $V_B([0, 1]) = \infty$ a.s.

Proof for Corollary.

■.

$$\begin{aligned}(\star) \sum_{k=1}^{2^n} \left(B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}}\right)^2 &\le \sup_{1\le k\le 2^n} \left|B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}}\right| \cdot \sum_{k=1}^{2^n} \left|B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}}\right| \\ &\le V_B([0, 1]) \cdot \sup_{1\le k\le 2^n} \left|B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}}\right|\end{aligned}$$

If $V_B([0, 1]) < \infty$, then $(\star) \to 0$.

Contradiction with $(\star) \to 1$, so w.p.1 $V_B([0, 1]) = \infty$. □

$(H \cdot X)_t = \int_0^t H_s\ \mathrm{d}X_s$, where H_s is previsible or predictable, X_s is semi martingale.

Example . Let $H_n \in \mathcal{F}_{n-1}$, (M_n) be a mart.

Then $(H \cdot M)_n = \sum_{k=1}^n H_k(M_k - M_{k-1})$ is a martingale.

1.1 Set-up

Given a probability space $(\Omega, \mathcal{F}, \mathsf{P})$. A stochastic process $X = (X_t)_{t\geq 0}$ is a collection of random variables taking values in $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$.

$$\begin{aligned} X\colon [0,\infty) \times \Omega &\to \mathbb{R}^d \\ (t,\omega) &\mapsto X_t(\omega). \end{aligned}$$

Fix ω, then the function $t \mapsto X_t(\omega)$ is called a sample path.

Definition 1.1.1. For two stochastic process X and Y, we say Y is a modification of X if

$$\forall\ t \geq 0, \quad \mathsf{P}(X_t = Y_t) = 1$$

Definition 1.1.2. We say X and Y are indistinguishable if

$$\mathsf{P}(X_t = Y_t,\ \forall\ t \geq 0) = 1.$$

And then Y is a version of X.

Example . Let Y be a modification of X and suppose X and Y both have a.s. right continuous sample paths. Then X & Y are indistinguishable.

Definition 1.1.3. A function f is right-continuous if $f(x) = f(x+) = \lim_{t\downarrow x} f(t)$.

Proof. Since $\forall\ t \geq 0$, $\mathsf{P}(X_t \neq Y_t) = 0$, we get

$$\mathsf{P}\left(\bigcup_{q\in\mathbb{Q}^+} X_q \neq Y_q\right) \leq \sum_{q\in\mathbb{Q}^+} \mathsf{P}(X_q \neq Y_q) = 0.$$

$$\Rightarrow\ \mathsf{P}\left(\bigcap_{q\in\mathbb{Q}^+} X_q = Y_q\right) = 1.$$

For two right-continuous functions f and g, if $f(q) = g(q)$, $\forall\ q \in \mathbb{Q}^+$, then $f(t) = g(t)$, $\forall\ t \geq 0$.

Hence $\mathsf{P}\left(X_t = Y_t,\ \forall\ t \geq 0\right) = 1$. □

Example . Let $T \sim \mathrm{Exp}(1)$. Define

$$\begin{aligned} X_t &\equiv 0, \quad \forall\ t \geq 0 \\ Y_t &= \begin{cases} 0, & t \neq T \\ 1, & t = T. \end{cases} \end{aligned}$$

$\forall\ t \geq 0$, $\mathsf{P}(X_t \neq Y_t) = \mathsf{P}(Y_t = 1) = \mathsf{P}(t = T) = 0$.

1.2 Filtration

Let $(\mathcal{F}_t)_{t\geq 0}$ be a filtration of $(\Omega, \mathcal{F})$ such that $\forall\ 0 \leq s < t < \infty$, $\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}$.

Example . (Natural Filtration) Let $(X_t)_{t\geq 0}$ be a stochastic process, Define

$$\mathcal{F}_t^X = \sigma(X_s, 0 \leq s \leq t).$$

Then $(\mathcal{F}_t^X)$ is a filtration.

Obviously, $X_t \in \mathcal{F}_t^X$.

Exercise 1.7. Let X be a process, every sample path of which is <u>RCLL</u>. Let

$$A = \{\omega \colon X(\omega) \text{ is continuous on } [0, t_0)\}$$

for some fixed $t_0 > 0$. Show that $A \in \mathcal{F}_{t_0}^X$.

<u>RCLL</u>: $f(x) = f(x+)$ and $f(x-)$ exists.

Proof.

$$\begin{aligned} A &= \{\omega \colon \forall\ s \in (0, t_0),\ X(s-, \omega) = X(s, \omega)\} \\ &= \{\omega \colon \forall\ q \in (0, t_0) \cap \mathbb{Q},\ X(q-, \omega) = X(q, \omega)\} \end{aligned}$$

Let $(0, t_0) \cap \mathbb{Q} = (q_m)_{m\geq 1}$, then

$$\begin{aligned} A &= \left\{\omega \colon \forall\ q_m \text{ with } m \geq 1,\ \forall\ k \geq 1,\ \exists\ N \geq 1,\ \forall\ n \geq N, \left|X(q_m - \frac{1}{n}, \omega) - X(q_m, \omega)\right| < \frac{1}{k}\right\} \\ &= \bigcap_{m=1}^{\infty} \bigcap_{k=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} \left\{\left|X(q_m - \frac{1}{n}, X(q_m)) < \frac{1}{k}\right|\right\} \in \mathcal{F}_{t_0}^X. \end{aligned}$$

□

Exercise 1.8. Let X be a process whose sample paths are RCLL a.s. Then $A \notin \mathcal{F}_{t_0}^X$ is possible.

Proof. [TODO] □

Definition 1.2.1. A filtration $\{\mathcal{F}_t\}_{t\geq 0}$ is said to satisfy the <u>usual condition</u> if

(a) $\mathcal{F}_t = \mathcal{F}_{t+} = \bigcap_{s>t} \mathcal{F}_s$;

(b) $\mathcal{F}_0$ contains all the <u>$\mathsf{P}-$negligible</u> sets in $\mathcal{F}$.

Definition 1.2.2. An even A is P-negligible if $\exists\ N \in \mathcal{F}$ s.t. $A \subseteq N$, $\mathsf{P}(N) = 0$.

Definition 1.2.3. $\mathcal{F}_{t+} = \bigcap_{s>t} \mathcal{F}_s$, $\mathcal{F}_{t-} = \sigma\left(\bigcup_{s<t} \mathcal{F}_s\right)$. We say a filtration is right-continuous if $\mathcal{F}_t = \mathcal{F}_{t+}$.

Definition 1.2.4. The stochastic process X is adapted to the filtration $\{\mathcal{F}_t\}_{t\geq 0}$ if $\forall\ t \geq 0$, $X_t \in \mathcal{F}_t$.

Obviously (X_t) is adapted to the natural filtration.

Definition 1.2.5. The stochastic process X is called progressively measurable with respect to $(\mathcal{F}_t)$ if $\forall\ t \geq 0$, the map

$$(s,\omega) \mapsto X_s(\omega)$$
$$([0,t] \times \Omega, \mathcal{B}([0,t]) \otimes \mathcal{F}_t) \to (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$$

is measurable.

adapted $\supseteq$ progressively measurable $\supseteq$ previsible

Proposition 1.2.1. If X is adapted to the filtration $(\mathcal{F}_t)$ and every sample path is right-continuous or left-continuous, then X is also progressively measurable.

Proof for Proposition.

■. Assume right-continuity $\forall\ t > 0$, $\forall\ n \geq 1$, $\forall\ 0 \leq k \leq 2^n - 1$,

$$X_s^{(n)}(\omega) = X_{\frac{k+1}{2^n}t}(\omega), \quad \forall\ \frac{kt}{2^n} < s \leq \frac{k+1}{2^n}t$$

Then $\lim\limits_{n\to\infty} X_s^n(\omega) = X_s(\omega)$ for all $(s,\omega) \in [0,t] \times \Omega$.

It suffices to prove

$$(s,\omega) \mapsto X_s^{(n)}(\omega)$$
$$([0,t] \times \Omega, \mathcal{B}([0,t]) \otimes \mathcal{F}_t) \to (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$$

is measurable.

$\forall\ A \in \mathcal{B}(\mathbb{R}^d)$,

$$\begin{aligned}&\left\{(s,\omega)\colon X_s^{(n)}(\omega) \in A\right\}\\ =&\bigcup_{k=0}^{2^n-1}\left(\frac{k}{2^n}t, \frac{k+1}{2^n}t\right] \times \left\{X_{\frac{k+1}{2^n}t} \in A\right\} \in \mathcal{B}([0,t]) \otimes \mathcal{F}_t\end{aligned}$$

We get $X_s^{(n)}(\omega)$ is measurable. Hence $X_s = \lim_{n\to\infty} X_s^{(n)}$ is also measurable. □

1.3 Stopping Times

$(\Omega, \mathcal{F}, \mathsf{P})$ and $(\mathcal{F}_t)_{t\geq 0}$.

Definition 1.3.1. A random time T is a stopping time of $(\mathcal{F}_t)$ if $\forall\ t \geq 0$, $\{T \leq t\} \in \mathcal{F}_t$.

Definition 1.3.2. A random time T is an optional time of $(\mathcal{F}_t)$ if $\forall\ t \geq 0$, $\{T < t\} \in \mathcal{F}_t$.

Proposition 1.3.1. stopping time $\subseteq$ optional time and stopping $=$ optional if the filtration is right-continuous.

Proof for Proposition.

- "$\Rightarrow$" $\{T < t\} = \bigcup_{n=1}^{\infty}\{T \le t - \frac{1}{n}\} \in \mathcal{F}_t$.
  "$\Leftarrow$" If T is an optional time, then $\forall\ m \ge 1$,

$$\{T \le t\} = \bigcap_{n=m}^{\infty}\left\{T < t + \frac{1}{n}\right\}.$$

Because $\{T < t + \frac{1}{n}\} \in \mathcal{F}_{t+\frac{1}{n}} \subseteq \mathcal{F}_{t+\frac{1}{m}}$.

$\Rightarrow\ \{T \le t\} \in \mathcal{F}_{t+\frac{1}{m}}$

$\Rightarrow\ \{T \le t\} \in \bigcap_{m=1}^{\infty}\mathcal{F}_{t+\frac{1}{m}} = \mathcal{F}_{t^+} = \mathcal{F}_t$. □

Lemma 1.3.1. If T and S are stopping times, then $T \wedge S$, $T \vee S$, $T + S$ are stopping times.

Proof. $\forall\ t \ge 0$,

1. $\{T \wedge S \le t\} = \{T \le t\} \cup \{S \le t\} \in \mathcal{F}_t$.
2. $\{T \vee S \le t\} = \{T \le t\} \cap \{S \le t\} \in \mathcal{F}_t$.
3. $\{T + S \le t\} \in \mathcal{F}_t \Leftrightarrow \{T + S > t\} \in \mathcal{F}_t$.

$$\begin{aligned}
&\{T + S > t\}\\
=&\{T = 0,\ S > t\} \cup \{S = 0,\ T > t\} \cup \{0 < T < t,\ T + S > t\} \cup \{S > 0,\ T \ge t\}\\
\in&\mathcal{F}_t \cup \mathcal{F}_t \cup \{0 < T < t,\ T + S > t\} \cup \mathcal{F}_t.
\end{aligned}$$

It suffices to show

$$\{0 < T < t,\ T + S > t\} = \bigcup_{\substack{r \in \mathbb{Q}^+\\ 0<r<t}} \{r < T < t,\ S > t - r\}.$$

Because the right term belong $\mathcal{F}_t$.

"$\Leftarrow$ " $S > t - r > t - T$, so $S + T > t$.

"$\Rightarrow$ " $\exists\ r \in (0, T)$, $S + r > t$ since $S + T > t$.

□

Lemma 1.3.2. If $\{T_n\}_{n\ge1}$ are stopping times, then $\sup_{n\ge1} T_n$ is a stopping time.

Proof. $\forall\ t \ge 0$, $\{\sup_{n\ge1} T_n \le t\} = \bigcap_{n=1}^{\infty}\{T_n \le t\} \in \mathcal{F}_t$. □

Definition 1.3.3. The stopping σ-algebra is $\mathcal{F}_T = \{A \in \mathcal{F} \mid \forall\ t \ge 0, A \cap \{T \le t\} \in \mathcal{F}_t\}$.

Proposition 1.3.2. If $T \equiv t_0$, then $\mathcal{F}_T = \mathcal{F}_{t_0}$.

Proof. $\forall\ t \geq 0$,

$$\{T \leq t\} = \begin{cases} \varnothing, & t_0 > t \\ \Omega, & t_0 \leq t \end{cases}$$

"$\mathcal{F}_{t_0} \subseteq \mathcal{F}_T$"

$\forall\ A \in \mathcal{F}_{t_0}, \forall\ t \geq 0$,

① $t \geq t_0$, $A \cap \{T \leq t\} = A \in \mathcal{F}_{t_0} \subseteq \mathcal{F}_t$;

② $t < t_0$, $A \cap \{T \leq t\} = \varnothing \in \mathcal{F}_t$.

Hence $A \in \mathcal{F}_T \Rightarrow\ \mathcal{F}_{t_0} \subseteq \mathcal{F}_T$.

"$\mathcal{F}_T \subseteq \mathcal{F}_{t_0}$"

$\forall\ A \in \mathcal{F}_T$, we have $A \cap \{T \leq t\} \in \mathcal{F}_t$ by definition. Let $t = t_0$, then $\{T \leq t_0\} = \Omega$, so

$$A \cap \{T \leq t_0\} \in \mathcal{F}_{t_0} \Rightarrow\ A \cap \Omega \in \mathcal{F}_{t_0} \Rightarrow\ A \in \mathcal{F}_{t_0}.$$

Hence $A \in \mathcal{F}_{t_0} \Rightarrow\ \mathcal{F}_T \subseteq \mathcal{F}_{t_0}$ $\square$

Definition 1.3.4 (defn example)**.** Example

Theorem 1.3.1 (thm example)**.** Example

Proof. proof □

Lemma 1.3.3 (lemma example)**.** Example

Proof for Lemma.

■. proof for lemma example □

Proposition 1.3.3. pro example

Proof for Proposition.

■. Example □

Claim 1.3.1. claim example Example

Proof for Claim.

■. proof for claim example □

Corollary 1.3.1 (cor example)**.** example

Proof for Corollary.

■. proof for corollary example □

Fact 1.3.1. Example fact

Remark. Example remark

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